

WEEK5 SELF STUDY ON COMPUTER SUPPORT AND MAINTENANCE

YEAR 1: SEMESTER 2

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1. Scenario Analysis : PC Build Description (150 words)

A typical PC assembly scenario involves core components of a computer. This includes the motherboard, CPU, RAM, power supply unit (PSU), storage devices (HDD/SSD), and GPU. First, the motherboard is mounted into the case, followed by installation of the CPU and applying thermal paste. RAM sticks are then inserted into their respective slots. The PSU is connected to the motherboard, GPU, and storage devices. Cables for peripherals are connected. Each component is secured properly to prevent damage during operation. And once everything is installed, the system is powered on to test functionality. This assembly requires proper knowledge of components compatibility, cable management, and troubleshooting.

2. Concept Research: Anti-Static Measures (100 words)

Anti-Static measures are essential for protecting computer components from static electricity, which can cause severe damage. The main techniques include using anti-static wrist straps, mats, and working on anti-static surfaces. An anti-static wrist strap is worn around the wrist and grounded to prevent static discharge onto sensitive parts. Anti-Static mats are placed on the work surface, providing a safe area for assembling parts. Keeping components in anti-static bags before installation further protects them from electrostatic discharge (ESD). These precautions ensure the longevity and stability of electronic components by minimizing the risk of short circuits or component failure due to static electricity.

3. Tool practice: PC Assembly Checklist (100 words)

- Ensure static –free workspace.
- Verify compatibility of all components (motherboard, CPU, RAM, PSU, GPU, and Storage).
- Install CPU onto motherboard, apply thermal paste.
- Install RAM onto appropriate slots.
- Mount motherboard into case and secure it.
- Install PSU and connect cables to motherboard, GPU, and storage devices.
- Insert storage devices (HDD/SSD) and connect data and ensure secure connection.

- Connect peripherals (keyboard, mouse, and monitor).
- Power on and check system functionality.
- Test for errors and verify performance.

Reflection:

The checklist serves as a clear guideline for assembling a PC. It helps avoid errors and ensures a smooth process by confirming the compatibility and proper installation of all components before powering up the system. The checklist makes assembly simpler and provide confidence in avoiding mishaps during the process.

4. Diagram Design :PC Assembly Diagram (100 words)

A well – designed PC assembly diagram visually illustrates the connections and involved in building a computer. The diagram typically highlights the CPU placement on the motherboard, RAM SLOT storage devices. It also includes the PSU' installation, power connections and storage devices. It also includes the PSU's wiring layout, showing how connects to each component for power. Using Canva, design a clear labels diagram simplifies the process, offering a reference guide for builders. It also helps ensure components are connected properly to avoid operational issues.

